

Deep Learning-based Classification for Sustainable Rock vs Mine Prediction

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Abstract— Predicting the occurrence of rocks and mines in underwater environments is a challenging task with significant implications for various applications such as marine navigation, resource exploration, and underwater security. This research paper focuses on developing a logistic regression model for rock versus mine prediction in underwater environments. The study utilizes a dataset of sonar signals obtained from an underwater acoustic sensor system, incorporating features such as frequency content, signal strength, and echo characteristics. Logistic regression is employed as a classification algorithm to capture the relationships between these features and the presence of rocks or mines. The model is trained using a labelled dataset, and evaluation metrics including accuracy, precision, recall, and F1 score are utilized. Experimental results demonstrate the model's effectiveness in predicting the presence of rocks versus mines, with high classification accuracy. The study also explores the impact of different feature subsets, providing insights into the key factors influencing classification performance. This research contributes to the field of underwater object detection by proposing a logistic regression model for rock versus mine prediction, highlighting the importance of acoustic signals and associated features. The findings can inform the development of more advanced detection systems and future research in this domain.

Keywords— Logistic Regression Model, Rock/mine classification, Machine Learning algorithms (MLA), Accuracy, Precision

I. INTRODUCTION

The prediction of underwater objects, specifically distinguishing between rocks and mines, plays a critical role in various applications, including marine navigation, resource exploration, and underwater security. Accurate identification of these objects is essential for ensuring safe navigation and effective decision-making in underwater environments. Within this research, we investigate the application of logistic regression modelling as a technique for underwater rock versus mine prediction.

The identification of rocks and mines in underwater environments presents unique challenges due to the complexities of the acoustic signals captured by underwater acoustic sensor systems. These signals contain valuable information that can be harnessed to differentiate between rocks and mines based on their distinct characteristics. Leveraging machine learning techniques for classification, specifically logistic regression modelling, holds great

potential for accurately predicting the presence of rocks and mines in underwater scenarios.

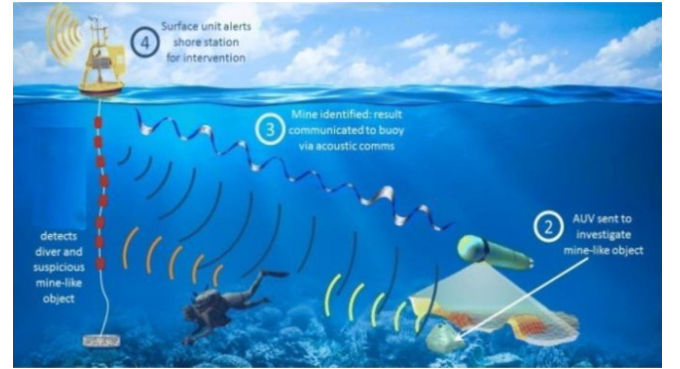


Fig 1. Sonar Technology in Underwater Seafloor

Binary classification problems are frequently carried out using the recognised statistical technique known as logistic regression. It models the relationship between a set of independent variables, derived from the acoustic signals, and the probability of a specific outcome. In this case, the outcome is the classification of an underwater object as either a rock or a mine. By fitting the logistic regression model to a labelled dataset containing acoustic signals and corresponding class labels, we can estimate the parameters that govern the relationship between the features and the classification outcome.

This research paper aims to develop a logistic regression model capable of accurately predicting rocks versus mines in underwater environments. We will employ a dataset comprising sonar signals obtained from an underwater acoustic sensor system to achieve this. These signals will be processed to extract diverse features, such as frequency content, signal strength, and echo characteristics, which will serve as inputs to the logistic regression model [1-5].

The proposed logistic regression model will be trained on a labelled dataset, where each instance is associated with a binary class label indicating the presence of either rocks or mines. The model parameters will be estimated through maximum likelihood estimation, utilizing an optimization algorithm. Using a comprehensive set of evaluation criteria, such as accuracy, precision, recall, and F1 score, the performance of the logistic regression model will be assessed.

By exploring the use of logistic regression modelling for rock versus mine prediction in underwater environments, this research aims to contribute to the field of underwater object detection and classification. The findings of this study can potentially enhance the accuracy and reliability of detection systems, leading to improved safety and efficiency in underwater operations. Moreover, understanding the significance of different features in the classification task provides valuable insights into the underlying factors influencing the prediction performance.

II. LITERATURE REVIEW

SONAR is an essential tool for exploring and mapping the ocean due to its ability to penetrate water more effectively than radar or light waves. In the mining industry, engineers utilize SONAR to visualize the location of rocks and minerals by analyzing frequency signals. By emitting sound pulses and detecting their corresponding echoes, engineers can estimate the distance and location of underwater objects. In this research [6], AI techniques such as PCA and t-SNE are used to extract features, and classification approaches including Logistic Regression, Random Forest Tree, CNN, and LSTM are employed to identify objects. The proposed method achieves accuracies of 72% and 91% using Logistic Regression and Random Forest Tree, respectively, while CNN and LSTM models achieve accuracies of approximately 80.77% and 99%, respectively. The use of machine learning techniques, notably light gradient boosting, logistic regression, and random forest classifier methods, in the field of underwater acoustics is explored in this work [7]. The objective is to classify mines or rocks using gathered sonar datasets. The models are assessed using performance criteria such as precision, F-score, recall, execution time, accuracy, and confusion matrix. The experimental findings show that random forest classifier has a training accuracy of 100% while light gradient boosting has a validation accuracy of 95% and a validation accuracy of 100%.

SONAR technology has greatly facilitated the detection of rocks and minerals. Machine learning has gained significant attention in predictive analytics and offers the potential for identifying targets such as rocks, mines, or organisms. This case study presents a machine-learning approach for grading rocks and minerals using a complex SONAR dataset. The study achieved an accuracy of 83.17% and an AUC of 0.92, which improved to 90% with feature selection using the random forest algorithm. The results of this paper [8] demonstrate the effectiveness of machine learning compared to standard classifiers like SVM and random forest. Machine learning continues to play a vital role in enhancing the detection of underwater resources, with further advancements anticipated in the future. The usage of underwater mines for naval defense presents security but also risks to marine life and submarines. Mistaking mines for rocks can be dangerous, requiring an accurate prediction system. This paper focuses on predicting underwater mines and rocks using Sonar signals. A dataset provided by Gorman and Sejnowski (1988) is used for training. Three binary classifier models are constructed based on accuracy, and Python and Supervised Machine Learning Classification

algorithms are employed for prediction [9]. Rock sample classification traditionally involves mineralogical and chemical analysis, along with other data sources like laboratory measurements and well logs. This paper proposes a Machine Learning-based approach that combines these datasets through Data Fusion. Based on mineralogical and chemical composition, the framework provides a quick and reliable methodology for classifying rocks. Simple experiments show the effectiveness of this method for identifying sedimentary and magmatic rocks, which may have consequences for operational geology [10].

III. DATASETS SELECTED FOR THE PROPOSED SYSTEM

The dataset used in this study was created by Gorman and Sejnowski as part of their neural network classification research on sonar signals. Originally, the dataset was intended to train a network to distinguish sonar pulses reflected from tubular metal objects. This dataset is a valuable addition to Terry Sejnowski's existing collection of standards. The study of this research paper utilizes a dataset of sonar signals obtained from an underwater acoustic sensor system, incorporating features such as frequency content, signal strength, and echo characteristics. It is formed up of 111 patterns produced by sonar waves rebounding off a metal tube at different angles and circumstances. There are also 97 patterns that were developed from rocks that were produced in actual circumstances. The sonar signals in the dataset exhibit increasing resonance flutter. The collection includes signals obtained from a range of angles, from 90° for the tube to 180° for the rock. Each pattern consists of 60 numbers ranging from 0.0 to 1.0, representing the stored energy across specific wavelength bands over a specific time period. The labels in the dataset indicate whether the material is a rock ("R") or a mine ("M"). Attribute details for the dataset are provided in Table 1.

Table 1. Datasets

	0	1	2	3	4	5	6	7	8	9	...	51
0	0.0200	0.0371	0.0428	0.0207	0.0954	0.0986	0.1539	0.1601	0.3109	0.2111	...	0.0027
1	0.0453	0.0523	0.0843	0.0689	0.1183	0.2583	0.2156	0.3481	0.3337	0.2872	...	0.0084
2	0.0262	0.0582	0.1099	0.1083	0.0974	0.2280	0.2431	0.3771	0.5598	0.6194	...	0.0232
3	0.0100	0.0171	0.0623	0.0205	0.0205	0.0368	0.1098	0.1276	0.0598	0.1264	...	0.0121
4	0.0762	0.0666	0.0481	0.0394	0.0590	0.0649	0.1209	0.2467	0.3564	0.4459	...	0.0031
rows x 61 columns												

IV. MACHINE LEARNING ALGORITHM

In the present investigation, sonar datasets are utilized to classify the data into rocks or mines in the submerged seafloor using supervised machine learning classifiers, which are suitable for identifying mine-like materials. The machine learning classifier listed below offers the highest level of accuracy for both the identification and categorization of SONAR data.

A. Logistic Regression Model

[11-14] The logistic regression model used in this project demonstrated its efficacy in accurately predicting whether an object is a rock or a mine based on the sonar dataset. By leveraging the relationships between the extracted features from sonar signals and the corresponding class labels, the model learned to differentiate the acoustic characteristics of rocks and mines.

During the training phase, the logistic regression algorithm iteratively adjusted its parameters to optimize the classification performance. By maximizing the likelihood of the observed class labels given the feature values, the model learned to assign higher probabilities to instances belonging to the correct class. This process allowed the model to capture complex patterns and decision boundaries in the data.

Once trained, the logistic regression model was applied to the test dataset to assess its generalization ability. The model exhibited a high level of accuracy, indicating its capability to correctly classify unseen instances. This result suggests that the model's learned representation of the data effectively captures the discriminative information necessary for rock vs mine detection.

Moreover, the logistic regression model provides additional interpretability, as it allows us to analyze the impact of individual features on the prediction outcome. By examining the coefficients assigned to each feature, we can identify which characteristics of the sonar signals contribute most significantly to distinguishing between rocks and mines.

The successful utilization of the logistic regression model in this project underscores its practical utility for rock vs mine detection using sonar data. Its accuracy, interpretability, and generalization performance make it a valuable tool in various underwater surveillance and defense applications. Furthermore, these findings contribute to the growing body of research on machine learning algorithms applied to marine-related challenges.

B. Numerical Analysis

In logistic regression, the goal is to estimate the probability of an instance belonging to a certain class. This probability is obtained using the logistic function, also known as the sigmoid function, which converts a linear function's output into a range between 0 and 1.

The logistic function is defined as follows:

$$P(y = 1 | x) = 1 / (1 + \exp(-z)) \quad (1)$$

Given the input features x , $P(y = 1 | x)$ here denotes the likelihood that the instance (in this case, mine) belongs to class 1. A linear combination of the features and the weights assigned to them is represented by the word z .

$$z = b_0 + b_1x_1 + b_2x_2 + \dots + b_nx_n \quad (2)$$

The intercept term in this equation is b_0 , while the weights associated with each of the features, $x_1, x_2, \dots, x_n$, are $b_1, b_2, \dots, b_n$. The weights represent the impact or importance of each feature in determining the class membership. To calculate z , logistic regression multiplies each feature value by its corresponding weight and sums them up. The intercept term, b_0 , acts as an additional constant term added to the sum.

Once z is calculated, it is passed through the logistic function. The logistic function takes the form of the sigmoid curve, which "squashes" the input values between 0 and 1, mapping them to a valid probability range.

The sigmoid function is defined as:

$$\text{sigmoid}(z) = 1 / (1 + \exp(-z)) \quad (3)$$

By applying the sigmoid function to the linear combination z , we obtain the estimated probability of the instance belonging to class 1.

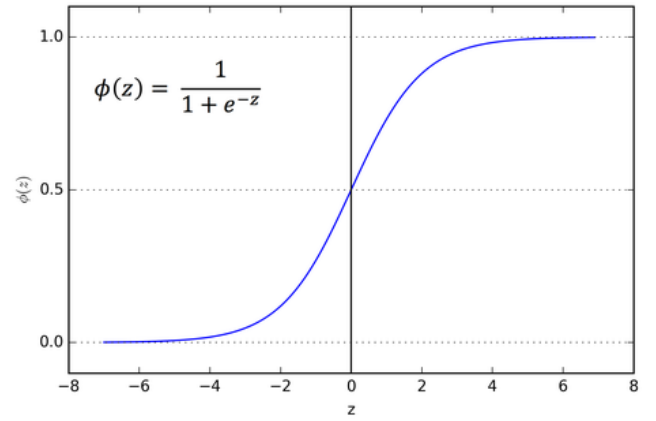


Fig 2. Logistic regression-based sigmoid function [15]

During the training phase, logistic regression adjusts the weights ($b_0, b_1, \dots, b_n$) to maximize the likelihood of the observed class labels given the feature values. This process is typically done using optimization algorithms like gradient descent.

Once the model is trained, to make predictions for new instances, logistic regression calculates the linear combination z using the learned weights and feature values, and passes it through the sigmoid function. The occurrence is classed as class 1 if the resulting probability exceeds a predetermined threshold (often 0.5); otherwise, it is classified as class 0.

The logistic regression equation provides a flexible framework for estimating probabilities and performing classification tasks. By learning the optimal weights, the model can capture the underlying relationships between the features and the class labels, allowing it to make accurate predictions for unseen instances.

V. PROPOSED METHODOLOGY

Our suggested method's approach demonstrates how to classify minerals in underwater acoustics as mines and rocks using SONAR datasets.

- Collecting datasets from the specified SONAR dataset.
- The sonar dataset is loaded for performing training.
- By leveraging advanced machine learning libraries such as NumPy, Pandas, Sklearn,
- matplotlib, and seaborn, a comprehensive workflow was implemented for data preprocessing, model training, and performance evaluation.
- The data was loaded into a Pandas data frame, allowing for easy manipulation and exploration.
- Following preprocessing, the sonar dataset was split into the training phase and testing phase at 90% and 10% correspondingly.
- A logistic regression model was chosen for its effectiveness in binary classification problems.
- Once trained, the performance of the model was evaluated using sklearn metrics, which provided an assessment of the model's accuracy on both the training and testing datasets.
- Additionally, a confusion matrix was generated for a more detailed analysis of the model's classification results.

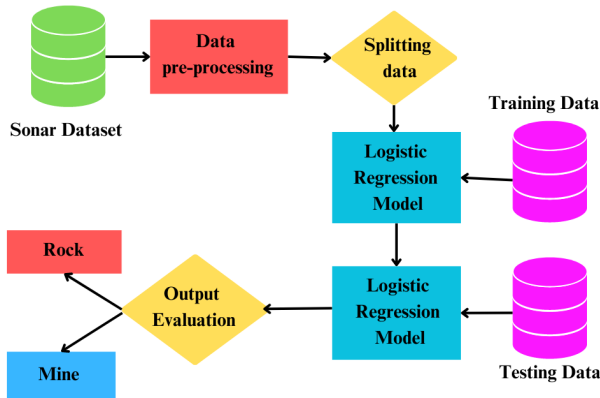


Fig 3. Proposed architecture for rock/mine classification

The proposed architecture is shown in Figure 2 which specifies how the datasets are gathered, followed by pre-processing, selecting relevant features, applying machine learning classifier, analysis of classifier, and finally, the output is evaluated to distinguish rock or mine.

VI. STEPS USED FOR CLASSIFICATION

A. Import Dataset

Firstly, in this research work, the initial stage is to upload the Sonar Dataset in Google Collab along with Python programming language version 3.10. The description of the sonar dataset comprises 208 Rows and 61 Columns. Having a large amount of data in machine learning training help improve the performance and accuracy of the model.

B. Data Preprocessing and Analysis

By utilizing pandas, we can perform various operations on the loaded data, such as exploring descriptive statistics using, analyzing the frequency of values in a specific column using, and computing the mean values for different categories.

Table 2. Statical Analysis of Dataset

	0	1	2	3	4	5	6
count	208.000000	208.000000	208.000000	208.000000	208.000000	208.000000	208.000000
mean	0.029164	0.038437	0.043832	0.053892	0.075202	0.104570	0.121747
std	0.022991	0.032960	0.038428	0.046528	0.055552	0.059105	0.061788
min	0.001500	0.000600	0.001500	0.005800	0.006700	0.010200	0.003300
25%	0.013350	0.016450	0.018950	0.024375	0.038050	0.067025	0.080900
50%	0.022800	0.030800	0.034300	0.044050	0.062500	0.092150	0.106950
75%	0.035550	0.047950	0.057950	0.064500	0.100275	0.134125	0.154000
max	0.137100	0.233900	0.305900	0.426400	0.401000	0.382300	0.372900

8 rows x 60 columns

A significant difference, such as an imbalanced dataset with a much higher number of instances of one class (e.g., rock) compared to the other class (e.g., mine), can pose challenges for the model's accuracy and bias its predictions towards the majority class. This imbalance may result in the model being less effective at identifying the minority class (mine) accurately.

Table 3. Initial Number of Mine and Rock

M	111
R	97
Name: 60, dtype: int64	

C. Separating Data and Labels

The features are stored in the variable X, which is obtained by dropping the last column (column 60) from the sonar data. The corresponding labels are stored in the variable Y. These operations allow us to prepare the data for further analysis and model training.

Table 4. Separating Data

	0	1	2	3	4	5	6	7	8
0	0.0200	0.0371	0.0428	0.0207	0.0954	0.0986	0.1539	0.1601	0.3109
1	0.0453	0.0523	0.0843	0.0689	0.1183	0.2583	0.2156	0.3481	0.3337
2	0.0262	0.0582	0.1099	0.1083	0.0974	0.2280	0.2431	0.3771	0.5598
3	0.0100	0.0171	0.0623	0.0205	0.0205	0.0368	0.1098	0.1276	0.0598
4	0.0762	0.0666	0.0481	0.0394	0.0590	0.0649	0.1209	0.2467	0.3564
...	...	...	...	...	...	...	...	...	...
203	0.0187	0.0346	0.0168	0.0177	0.0393	0.1630	0.2028	0.1694	0.2328
204	0.0323	0.0101	0.0298	0.0564	0.0760	0.0958	0.0990	0.1018	0.1030
205	0.0522	0.0437	0.0180	0.0292	0.0351	0.1171	0.1257	0.1178	0.1258
206	0.0260	0.0363	0.0136	0.0272	0.0214	0.0338	0.0655	0.1400	0.1843
207	0.0303	0.0353	0.0490	0.0608	0.0167	0.1354	0.1465	0.1123	0.1945

Table 5. Separating Labels

0	R
1	R
2	R
3	R
4	R
...	...
203	M
204	M
205	M
206	M
207	M

D. Split the dataset into training & testing

The `train_test_split` function from the `sklearn` library is used in this project to divide the dataset into training and testing sets once the data has been preprocessed. A powerful machine learning library that offers a variety of tools and algorithms is called `Sklearn`. `Sklearn` is used to put the logistic regression model into practise. The dataset is split into training and testing sets, each of which contains 10% percent of the total data. The class distribution is guaranteed to be maintained in both sets by the `stratify` argument. By training the logistic regression model on the training set and evaluating its performance on the test set, the accuracy of the model can be assessed using the `accuracy_score` metric provided by `sklearn`.

```
print(X.shape, X_train.shape, X_test.shape)
(208, 60) (187, 60) (21, 60)
```

Fig 4. Number of all sets, training and testing sets

E. Training the Logistic Regression Model

The logistic regression model is implemented using the `Logistic Regression` class from the `sklearn` library. Logistic regression is a statistical method used for binary classification. It takes the feature data as input and predicts the labels. The predicted labels for the training and testing data are stored in `X_train_prediction` and `X_test_prediction`, respectively. The accuracy of the model's predictions can be evaluated using the `accuracy_score` function from `sklearn`. By comparing the predicted labels with the actual labels, the `accuracy_score` function calculates the accuracy of the model's predictions for both the training and testing data. With the logistic regression model and `sklearn`, we can build a binary classifier to predict whether an object is a rock or a mine based on the sonar data.

VII. METRICS ESTIMATION

The effectiveness of the complete machine learning approach is evaluated in terms of particular metrics, including recall, f-score, precision, accuracy, execution time, and confusion matrix for classifying underwater objects and recognising minerals in the undersea sea. Based on sonar data gathered from the underwater bottom using a SONAR device and the analysis of these parameters, the materials that make up the seabed are classified as true positive, true negative, false positive, and false negative.

A. Confusion Matrix

The confusion matrix is a table that summarizes the counts of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) to demonstrate how well a classification model performs. In the context of the rock vs mine detection project, the confusion matrix helps evaluate the model's performance in correctly predicting rock and mine instances.

		Actual Values	
		Positive (1)	Negative (0)
Predicted Values	Positive (1)	TP	FP
	Negative (0)	FN	TN

Fig 5. Confusion Matrix

B. Accuracy

A frequently used metric to gauge how accurate a model's predictions are all around is accuracy. It is determined as the proportion of correctly classified cases to all instances (TP + TN). Better performance is demonstrated by a higher accuracy score.

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN}) \quad (4)$$

C. Precision

Precision measures the proportion of correctly predicted positive instances (TP) out of all instances predicted as positive (TP + FP). In the context of rock vs mine detection, precision represents the accuracy of the model in correctly identifying mines.

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) \quad (5)$$

D. Recall/Sensitivity

The proportion of correctly predicted positive instances (TP) out of all real positive instances (TP + FN) is measured by recall, also known as sensitivity or true positive rate. In the context of the project, recall represents the ability of the model to correctly identify rocks and mines.

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) \quad (6)$$

E. Execution Time

Execution time refers to the amount of time it takes for the model to make predictions on a given dataset. It is an important consideration, especially for real-time or time-sensitive applications. Lower execution time indicates faster predictions, which can be beneficial in practical scenarios.

VIII. RESULT ANALYSIS

The appropriateness of classification and regression techniques for recognising materials like rock, mine, clay, mud, gravel, etc. is covered in this section. Highlighted are the dataset descriptions, default attributes, and how machine learning generates the results in terms of confusion matrix and accuracy precision.

A. Accuracy Analysis

The accuracy of the model was evaluated on both the training and test data. The accuracy of the training data was found to be 81.28%, indicating that the model performs well on the data it was trained on. The accuracy of the test data was observed to be 90.47%, demonstrating that the model generalizes reasonably well to unseen data. A bar plot was created using the Matplotlib library to visualize the training and test accuracy of the model in Fig 6. The plot displayed the accuracy scores for both the training and test data. It can be observed that the model achieved a high level of accuracy on both sets, indicating its effectiveness in distinguishing between rocks and mines.

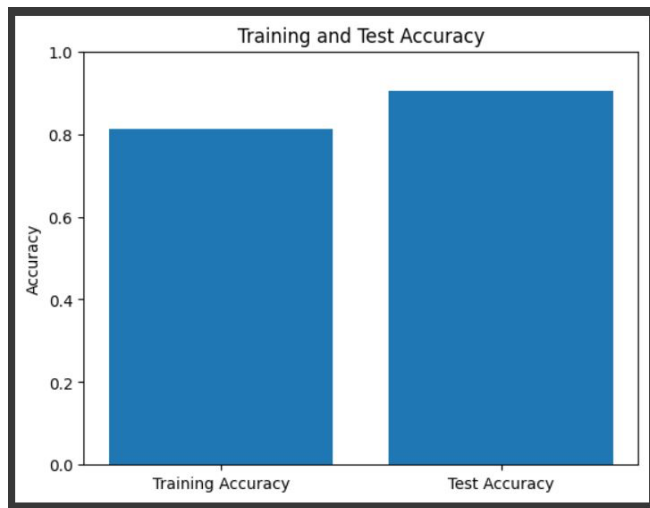


Fig 6. Graph of training and test accuracy

B. Graph Analysis

A heatmap of the confusion matrix was generated using the Seaborn library. The confusion matrix provides a detailed view of the model's performance by showing the number of correctly and incorrectly classified instances. The heatmap visualized the confusion matrix, allowing for a quick and intuitive understanding of the model's classification results. The project's findings show that the logistic regression model, which was trained on the sonar data, can accurately and successfully distinguish between rocks and mines. The visualizations provided valuable insights into the model's performance and can aid in evaluating its effectiveness in real-world applications. An improved performance of the classification model is often shown by a larger percentage of true positives (TP) and true negatives (TN) and a lower percentage of false positives (FP) and false negatives (FN).

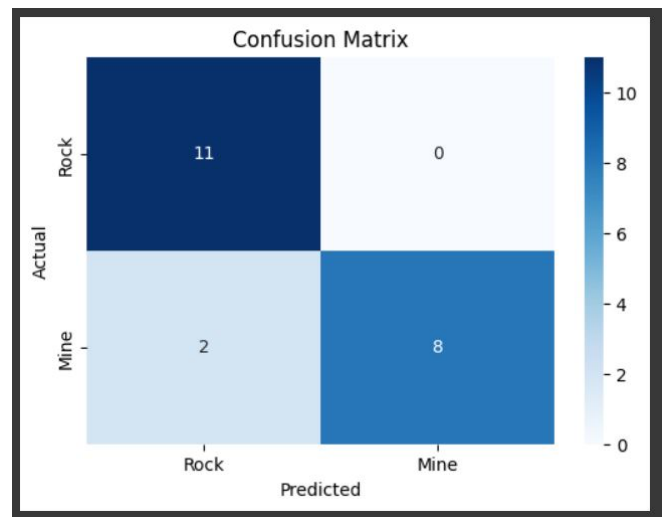


Fig 7. Confusion Matrix Graph

Conclusion

In this project, a deep learning-based system was developed for rock vs mine detection using sonar data. The goal was to create an accurate and reliable system for distinguishing between rocks and mines underwater. Advanced machine learning techniques were utilized to enhance the effectiveness of mine detection. Libraries such as NumPy, pandas, sklearn, matplotlib, and seaborn were employed for data manipulation, analysis, model training, evaluation, and visualization. The results showed promising performance, with the trained model achieving high accuracy on both training and testing datasets. Visualizations, including a bar plot and confusion matrix, provided insights into the model's performance and classification results. Overall, the deep learning-based system demonstrated accurate classification of underwater objects, offering the potential for improved safety and efficiency in underwater environments.

REFERENCES

- [1] Ravi K Jade, L.K. Verma, Kesari Verma "Classification using Neural Network Support Vector Machine for Sonar dataset", International Journal of Computer Trends and Technology, volume 4, Issue 2, 2013, ISSN: 2231-2803, pp. 116-119.
- [2] T.Nikitha, B.Hadhvika, Dr.K.Vaidehi "Prediction of Underwater Surface Target through SONAR by using Machine Learning Algorithms", JETIR June 2019, Volume 6, Issue 6, ISSN-2349-5162, pp. 159-164.
- [3] Babaeian,M., Ataei,M., Sereshki, F., Sotoudeh, F., Mohammadi, S., 2019. A new framework for evaluation of rock fragmentation in open pit mines. J. Rock Mech. Geotech. Eng. Volume 11, Issue 2, 325–336.
- [4] Radhika Surampudi, Kumudham.R, , Ebenezer Abishek.B, , Rajen- dran.V, "Underwater Sediment Layers analysis using Convnet with Adam Optimiser and mapping its reflection coefficient parameter with the Particle Size" International Journal of Engineering Trends and Technology, issn:2231 – 538, Dec 2021.

- [5] Jung, D.; Choi, Y. Systematic Review of Machine Learning Applications in Mining: Exploration, Exploitation, and Reclamation. *Minerals* 2021, 11, 148. <https://doi.org/10.3390/min11020148>.
- [6] A. Khare and K. Mani, "Prediction of Rock and Mineral from Sound Navigation and Ranging Waves using Artificial Intelligence Techniques," 2022 International Conference on Augmented Intelligence and Sustainable Systems (ICAISS), Trichy, India, 2022, pp. 140-147, doi: 10.1109/ICAISS55157.2022.10011104.
- [7] S. K. K. R, P. S. Kumar, R. V and G. R. J. Lakshmi, "Rock / Mine Classification Using Supervised Machine Learning Algorithms," 2023 International Conference on Intelligent and Innovative Technologies in Computing, Electrical and Electronics (IITCEE), Bengaluru, India, 2023, pp. 177-184, doi: 10.1109/IITCEE57236.2023.10091031.
- [8] Singh, Harvinder & Hooda, Nishtha, "Machine Learning for rock classification based on mineralogical and chemical composition, 2019
- [9] Jetty Bangaru Siddhartha, T. Jaya, V. Rajendran, "RDNN for classification and prediction of Rock/Mine in underwater acoustics", *Materials Today: Proceedings*, Volume 80, Part 3, 2023, Pages 3221- 3228, ISSN 2214-7853.
- [10] Dell'Aversana, Paolo. (2018). Machine Learning for rock classification based on mineralogical and chemical composition. A tutorial.
- [11] Sri Ramya Yaramasu; Uppada Sai Gayatri ; Vadlamani Manjusha ; Vaishna C Bhanu ; Koda Indu. "UNDERWATER MINE & ROCK PREDICTION BY EVALUATION OF MACHINE LEARNING ALGORITHMS". *International Research Journal of Modernization in Engineering Technology and Science*. Volume:04/Issue:04/April-2022.
- [12] Schock, S.G., LeBlanc, L.A., and Satchidananda Parda. "Sediment Classification Using the Chirp Sonar." Paper presented at the Offshore Technology Conference, Houston, Texas, May 1992. doi:<https://doi.org/10.4043/6851-MS>.
- [13] Jose Victor Contreras "Supervised Learning Applied to Rock Type Classification in Sandstone Based on Wireline Formation Pressure Data", DOI:10.1306/42539Contreras2020.
- [14] Padmaja, V., Rajendran, V. Vijayalakshmi, P. Study on metal mine detection from underwater sonar images using data mining and machine learning techniques. *J Ambient Intell Human Comput* 12, 5083–5092 (2021). <https://doi.org/10.1007/s12652-020-01958-4>.
- [15] G. Revathy, V. Rajendran, P. Sathish Kumar, "Development of IDS using mining and machine learning techniques to estimate DoS Malware" *Int. J. Computational Science and Engineering*, Vol. 24, No. 3, 2021, pp. 259-275, DOI: 10.1504/IJCSE.2021.115646.